

Microwave Engineering

Mid Sem (Lab Exam)

Set-A

Total Marks: 60

Time: 40 minutes

Instructions: Please tick the right option. There will be a 25% negative marking for multiple-choice questions.

Name:

Student ID:

1. In a microwave experiment, the total transmitted power is measured as 75% of the input power to the device. Calculate the magnitude of the reflection coefficient (normal scale). [2]

- a) 0.5 b) 0.25 c) 0.05 d) 0.025

2. How would you measure VSWR precisely through DSO? [1]

a) using V_{peak} to peak

b) using V_{max} to V_{min}

c) using V_{max} only

d) using V_{min} only

3. Which electron, while characterising a Klystron, travels the most in the Repeller space? [2]

- a) Reference electron b) Late electron c) Early electron d) All of the above

4. The microwave bench setup commonly used in our lab (with a reflex klystron as source) operates in which frequency band? [1]

- a) X-Band b) C-Band c) S-Band d) L-Band

5. What does 'CW' mentioned in the Klystron power supply stand for? [1]

- a) Continuous Wave b) Compelling Wave c) Constant Wave d) Coherent Wave

6. Which one of the following modes is the dominant mode? [2]

- a) $3\lambda/4$ b) $5\lambda/4$ c) $\lambda/4$ d) $\lambda/2$

7. What is the least count value (cm) used while making the calculation using the slotted wave guide section's scale? [2]

- a) 0.1 b) 0.001 c) 0.01 d) none

8. Circular Cavities are used for microwave frequency meters. The cavity is constructed with a movable top wall to allow _____ of resonant frequency. [2]

- a) tuning b) calibrating c) stabilizing d) both a & b

9. What is the application of a characterised Klystron tube used in the lab? [2]

- a) Amplifier b) Oscillator c) RF source d) both b & c e) both a & c

10. What is the full form of CVD? [1]

- a) Conflicting Vernier Dimension
b) Cohering Vernier Dimension

c) Combined Vernier Dimension

d) Coinciding Vernier Dimension

11. What should be the initial position for the beam voltage while turning on the klystron tube power supply? [2]

- a) maximum b) minimum c) Switched off d) None

12. In a microwave bench using a reflex klystron, which part of the observed signal is mainly affected when the beam voltage is increased? [1]

- a) Amplitude b) Phase c) Frequency d) Power

13. A slotted line is a line configuration that allows the _____ of the amplitude of a standing wave on a terminated line, usually having a characteristic impedance of $50\ \Omega$. [2]

- a) synthesizing b) sectioning c) sampling d) segregating

14. While determining frequency, a user had to have _____ to obtain X_1 & X_2 to complete the calculation. [2]

- a) two consecutive minima
b) two consecutive maxima
c) either a or b
d) any two maxima

15. Suppose, at 4 GHz, the S_{11} is -20 dB. This is equivalent to a reflection coefficient of 0.1 in natural value at port 1. What is the power reflected (in percentage) from port one and transmitted to port 2? ____1%____ and ____99%____. (Assume the system is lossless) (True) [2]

16. In CST, mesh refinement is important because: [1]

- a) It reduces simulation time b) It increases accuracy of results c) It reduces memory requirement d) It avoids solver errors only

17. The component used to prevent power from reflecting back into the source is called a ____ Isolator _____. (True) [1]

18. The device used to measure VSWR directly is _____. [2]

- a) Slotted line (b) Klystron tube (c) Isolator (d) Spectrum analyzer

19. The operating modes of a reflex klystron are determined by the _____ voltage. [2]

- a) Beam b) Heater c) Repeller d) Modulator

20. While simulating on CST, if VSWR at a particular frequency is 1, then the value of S_{21} parameter (dB) should be _____. [2]

- a) 0 b) 1 c) -130 d) Infinity e) None of these

21. The dielectric constant of a material is determined by measuring the shift in _____. [2]

- a) Frequency b) Voltage c) Wavelength d) Attenuation

22. The cutoff frequency of a rectangular waveguide depends mainly on _____. [2]

a) Length of waveguide **b) Cross-sectional dimensions** c) Operating voltage d) Dielectric loss tangent

23. In CST, the return loss at port 1, the parameter is represented by _____. [2]

a) S21 **b) S11** c) S22 d) S12

24. For good antenna matching, S11 should be less than _____ dB. [2]

a) -1 b) -5 **c) -10** d) -50

25. The position of the stub is adjusted to cancel the _____ part of impedance. [2]

a) Resistive **b) Reactive** c) Both d) None

26. The transmission coefficient is represented in CST by _____. [2]

a) S11 b) S22 **c) S21** d) Γ

27. The dielectric constant of a typical FR4 substrate is around: [2]

a) 2.2
b) 4.4
c) 6.8
d) 10.2

28. The characteristic impedance of a microstrip line depends on: [2]

a) Strip width and substrate height
b) substrate width
c) strip height
d) None of these

29. Open boundary condition simulates: [2]

a) Free space b) Coaxial line c) Waveguide d) Resonator

30. Distance between two minima in terms of λ in the standing wave? [2]

a) λ b) $\lambda/4$ c) $\lambda \frac{3}{4}$ **d) $\lambda/2$**

31. While doing the CST simulation for a single stub matching, the stub position for a perfect matching should be: [2]

a) Towards the source **b) Towards the Load** c) Either is fine from the previous two options d) none of these

32. The value of the S11 parameter obtained in the CST simulation for a short-circuited and open-circuited line will be: [2]

a) both closer to 0 dB b) both closer to 1 dB c) 0 dB and 1 dB d) 1dB and 0 dB e) -40 dB

33. The distance between the position of $V_{\max}$ and $V_{\min}$ is : [1]

a) λ **b) $\lambda/4$** c) $\lambda \frac{3}{4}$ d) $\lambda/2$

34. While doing the experiment on the Test bench, we were using the following components:

Isolator (1), Klystron tube (2), frequency meter (3), attenuator (4), detector mount (5), and slotted line (6)

The correct order for placing the components on the test bench is:

[2]

a) 2-4-1-3-6-5

b) 2-1-4-3-6-5

c) 2-4-1-3-5-6

d) 2-1-3-4-5-6

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a) $3\lambda \frac{3}{4}$

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- a) both closer to 0 dB** b) both closer to 1 dB c) 0 dB and 1 dB d) 1dB and 0 dB e) -40 dB

32. The distance between the position of Vmax and Vmin is : [1]

- a) λ **b) $\lambda/4$** c) $\lambda \frac{3}{4}$ d) $\lambda/2$

33. Distance between two minima in terms of λ in the standing wave? [2]

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